



Probability I

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Stat 141 | Fall 2026
Monday, Week 3

Goals For Today

- Introduce **probability theory**
- See the **Law of Large Numbers**
- Define and practice using key **rules of probability**
- Introduce **conditional probability**

Probability Theory

Random Processes and Events

Probability Theory: the study and quantification of uncertainty/randomness in outcomes of repeated experiments.

A **random process** is one which we know what results *could* happen, but don't know which particular result *will* happen.

- Can be used to model processes that are complicated, but not truly random, to figure out how they work
- Example: Rolling a 6-sided die

An **event** is a potential result of some particular random process.

- Example: If we roll a 6-sided die, one event is "*the die rolls a 6*". Another is "*the die rolls an odd number*".
- The statement "*it will rain at 9am in Portland on 9/14*" is an event, but not one for the die-rolling process.

Perception of Probability

We will soon define the word “probability.” Before that, consider the following questions in small groups:

1. When flipping a fair coin, we might say that “the probability of flipping Heads is 0.5.” How do you interpret this sentence?
 - a. If I flip this coin over and over, roughly 50% will be Heads.
 - b. Heads and Tails are equally plausible.
 - c. Both a and b make sense.
2. An election is coming up and a pollster claims “candidate A has a 0.9 probability of winning”. How do you interpret this sentence?
 - a. If we observe the election over and over, candidate A will win roughly 90% of the time.
 - b. Candidate A is much more likely to win than to lose.
 - c. Both a and b make sense.

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Perception of Probability (Notes)

- If you answered (a), you may naturally subscribe to the “**Frequentist**” view of probability (*long-run proportion*)
- If you answered (b), you may naturally subscribe to the “**Bayesian**” view of probability (*relative chance*)
- If you answered (c), you fall somewhere in between!

Neither definition is “correct”. Statisticians are divided on how to interpret probability.

- In this class, we’ll discuss frequentist probability (see next slide)

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Probability

The “frequentist” definition of probability is...

Definition: Probability

The **probability** of a particular event is the proportion of times the event would occur, if we observed the random process an *arbitrarily large* number of times.

To say that *a coin has 50% probability of landing heads*, means that...

- We expect the proportion of heads in a large number of coin flips to be close to 0.5

Since probabilities are defined as a proportion, **they will always be values between 0 and 1**.

For brevity, we’ll represent statements like *the probability of the event “the coin lands heads” is 50%* using the notation:

$$P(\text{Heads}) = 0.5 \quad \text{or} \quad P(H) = 0.5$$

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Law of Large Numbers

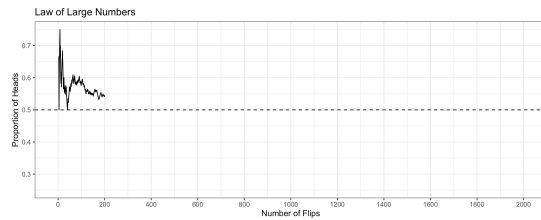
- **Q:** If I flip a fair coin 10 times, do we expect the proportion of heads observed to be 0.5?
- **Q:** What if I flip it 1000 times?
- **Q:** What do you expect to happen to the proportion of heads as we increase the number of flips?

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Law of Large Numbers

The probability of an event refers to the **long-run tendency of the proportion**.

- Observed proportions can deviate from expected probabilities in a finite number of trials
- But as you conduct more and more trials, the deviation decreases



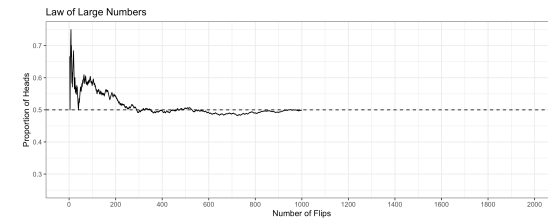
The proportion of heads deviates wildly from 0.5 during the first 200 flips

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Law of Large Numbers

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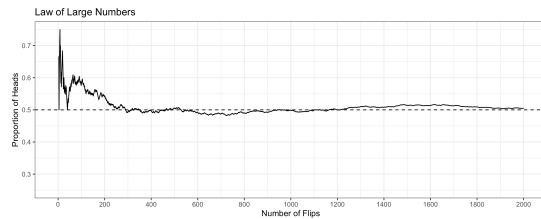
The proportion of heads gets closer to 0.5 after 1000 flips

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Law of Large Numbers

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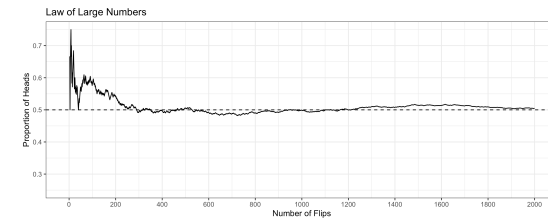
The proportion of heads is very close to 0.5 after 2000 flips!

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Law of Large Numbers

The probability of an event refers to the **long-run tendency of the proportion**.

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- But as you conduct more and more trials, the deviation decreases



This is the **Law of Large Numbers** in effect!

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Check Your Understanding

Suppose you're interested in winning the Powerball Jackpot, where the chance of winning is 1 in 292 million. You buy a single lottery ticket and wait to see if you won.

1. When playing the lottery (a *random process*), what are the two possible outcomes (**events**)?
2. Write down the probability of your two events.
3. If you played the lottery 1 million times, should you **expect** to win? What about 292 million times?
4. If you played the lottery 1 billion times, are you **guaranteed** to win?

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Rules of Probability

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Probability Models

We can formalize the connection between random processes/events with probabilities using a **Probability Model**

A **probability model** has two components:

1. A list of the possible results of a random process (called **events**)
2. A rule (called the **probability function**) that assigns to each event a probability between 0 and 1, in a consistent manner.

Example: Probability Model for a Coin Toss:

1. Heads or Tails
2. $P(\text{Heads}) = 0.5$ and $P(\text{Tails}) = 0.5$

When discussing probability, we always (explicitly or implicitly) define a probability model.

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Definition: Disjoint Events

Two events A and B are said to be **mutually exclusive** (or **disjoint**) if it is not possible for both to occur at the same time.

Example: Single Roll of a Die

- Let A denote the event "a 1 is rolled on 6-sided die"
- Let B denote the event "a 2 is rolled on 6-sided die"
- A and B are disjoint events for the random process

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The Addition Rule

Theorem: Addition Rule

The probability that at least one event occurs in a pair of disjoint events is the sum of their individual probabilities:

$$P(A \text{ or } B) = P(A) + P(B)$$

Q: When a die is rolled, what is the probability that an **odd number** is rolled?

$$P(\text{Odd}) = P(\text{roll a 1, 3, or 5})$$

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The Addition Rule

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Q: When a die is rolled, what is the probability that an **odd number** is rolled?

$$\begin{aligned} P(\text{Odd}) &= P(\text{roll a 1, 3, or 5}) \\ &= P(\text{roll a 1}) + P(\text{roll a 3}) + P(\text{roll a 5}) \quad (\text{by the Addition Rule}) \end{aligned}$$

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The Addition Rule

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The Addition Rule

Theorem: Addition Rule

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The Addition Rule: Only for Disjoint events!

Theorem: Addition Rule

The probability that at least one event occurs in a pair of disjoint events is the sum of their individual probabilities:

$$P(A \text{ or } B) = P(A) + P(B)$$

Q: In a die-roll, what's the probability we get an even number?

- $P(\text{Even}) = P(\text{Roll a 2, 4, or 6}) = \frac{3}{6} = \frac{1}{2}$

Q: In a die-roll, what's the probability we get at least a 3?

- $P(\geq 3) = P(\text{Roll a 3, 4, 5, or 6}) = \frac{4}{6} = \frac{2}{3}$

Q: What's the probability you get an even number OR at least a 3?

- $P(\text{Even OR } \geq 3) = P(\text{Roll a 2, 3, 4, 5, 6}) = \frac{5}{6}$

- But $P(\text{Even}) + P(\geq 3) = \frac{1}{2} + \frac{2}{3} = \frac{7}{6} \dots$ that's not even between 0 and 1!

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Complementary Events

The **complement** to an event A (denoted A^c) is the event that occurs exactly when the original does not.

- If A is the event that a 1 is rolled on a die, then A^c is the event that any other number (2,3,4,5, or 6) is rolled.

Theorem: Complement Rule

The probability that the complement of an event occurs is 1 minus the probability of the event:

$$P(A^c) = 1 - P(A)$$

What is the probability that any number other than a 1 is rolled on a fair 6-sided die?

$$P(\text{roll something other than a 1}) = 1 - P(\text{roll a 1}) = 1 - \frac{1}{6} = \frac{5}{6}$$

- By definition, complementary events are mutually exclusive

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Check Your Understanding

Theorem: Addition Rule

The probability that at least one event occurs in a pair of disjoint events is the sum of their individual probabilities:

$$P(A \text{ or } B) = P(A) + P(B)$$

Theorem: Complement Rule

The probability that the complement of an event occurs is 1 minus the probability of the event:

$$P(A^c) = 1 - P(A)$$

In the city of Portland, there's rain on 60% of days and snow on 1% of days.

- On a given day, what's the probability that it **doesn't rain**?
- On a given day, what's the probability that it rains **OR** doesn't rain?
- What if I told you it rains OR snows 61% of days in Portland. What's the **flaw** in my reasoning?

02:00

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Conditional Probability

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Activity: Coffee or Tea?

A survey was given to 100 Stat 141 students in 2017. Some results are summarized below:

	Coffee	Tea	total
First-year	7	10	17
Sophomore	25	20	45
Junior	13	12	25
Senior	8	5	13
total	53	47	100

1. What is the probability that a **random student** prefers *coffee*?
2. What is the probability that a **random student** was a *sophomore*?
3. What is the probability that a **random student** was a *sophomore* and preferred *coffee*?
4. What is the probability that a **random sophomore** preferred *coffee*?

03:00

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Conditional Probability

Conditional Probability: probability of something occurring, **given** that another event has **already** occurred.

We write *the conditional probability of Event A given Event B has occurred* as,

$$P(A | B)$$

In the previous example,

- Question: What is the probability that a random *sophomore* preferred *coffee*?
- Event A: The student prefers coffee
- Event B: The student is a sophomore
- Answer: $P(\text{Coffee} | \text{Sophomore}) = P(A|B)$

We calculated this intuitively before, but we can generalize.

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Conditional Probability

How do we calculate conditional probabilities?

In general,

Theorem: Conditional Probability Rule

The Conditional Probability of an event A given another event B is

$$P(A|B) = \frac{P(A \text{ and } B)}{P(B)}$$

$$P(\text{Coffee}|\text{Sophomore}) = \frac{P(\text{Sophomore and prefers Coffee})}{P(\text{Sophomore})}$$

- The probability that a student who is a sophomore (B) prefers coffee (A):

$$P(A | B) = \frac{P(A \text{ and } B)}{P(B)} = \frac{25}{100} \bigg/ \frac{45}{100} = \frac{25}{45}$$

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Recap

Definition: Probability

The **probability** of a particular **event** is the proportion of times the event would occur, if we observed the random process an *arbitrarily large* number of times.

Law of Large Numbers

Observed proportions may deviate from expected probabilities when n is small, but the deviation decreases as n increases. Think about the proportion of heads you expect from 10 vs 1,000,000 coin flips.

Theorem: Addition Rule

The probability that at least one event occurs in a pair of **disjoint** events is the sum of their individual probabilities:

$$P(A \text{ or } B) = P(A) + P(B)$$

Theorem: Complement Rule

$$P(A^c) = 1 - P(A)$$

Theorem: Conditional Probability Rule

The Conditional Probability of an event A given another event B is

$$P(A|B) = \frac{P(A \text{ and } B)}{P(B)}$$

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Next time

- More on conditional probability
- More probability rules
- More probability rule practice with activities

Quick note

- Even if you arrive to class late, you can fill out the attendance form by coming up after class!